

Assignment Week 5

Total marks=20

Q1. The unit of specific conductance (κ) is

- (a) $\Omega^{-1} \text{ cm}^{-1}$ (b) $\Omega^{-1} \text{ cm}^2$ (c) $\Omega \text{ cm}^{-1}$ (d) cm^{-1}

(1)

Q2. In a conductivity cell, the cell-constant depends on:

(a) distance between electrodes and electrode area

(b) electrolyte concentration

(c) nature of the ions present

(d) temperature only

(2)

Q3. Calculate the transport number of Na^+ and Cl^- ions when a current flows through an infinitely dilute aqueous solution of NaCl at room temperature (25 °C). Given the ionic mobilities (at infinite dilution) of $\text{Na}^+ = 5.20 \times 10^{-8} \text{ m}^2 \text{ V}^{-1} \text{ s}^{-1}$, and $\text{Cl}^- = 7.91 \times 10^{-8} \text{ m}^2 \text{ V}^{-1} \text{ s}^{-1}$.

a) 0.450, 0.550

b) 0.603, 0.397

c) 0.397, 0.603

d) 0.550, 0.450

$$t_+ = \frac{u_+}{u_+ + u_-}$$

(2)

Q4. The conductance of a solution is directly proportional to

(a) number of ions in solution

(b) ionic mobility

(c) temperature (for a given solution)

(d) all of the above

(2)

Q5. According to Friedrich Kohlrausch's law, at infinite dilution, the molar conductance of an electrolyte equals

(a) sum of ionic mobilities of the ions

(b) sum of limiting molar conductivities of cation and anion

(d) independent of ions

(2)

Q6. A decinormal solution of silver nitrate was electrolyzed between platinum electrodes. After passing current for 5 hours, a fall in concentration of 0.0007200 g-equivalent was observed in the anodic solution. The mass of copper deposited in a copper coulometer placed in series was found to be 0.05600 g. (Equivalent mass of Cu = 31.8) What is the transport number of the silver cation (Ag^+)?

a) 0.6102

b) 0.3898

c) 0.5904

d) 0.4090

$$q \text{ of Cu deposited} = \frac{\text{Mass}}{\text{Eq. Wt.}} = \frac{0.05600}{31.8} = 0.001761$$

$$t_{\text{Ag}^+} = \frac{\text{Loss of Ag}^+ \text{ in anode}}{\text{Total } q \text{ passed}}$$

$$t_{\text{Ag}^+} = \frac{0.000720}{0.001761} = 0.4090 \quad (2)$$

Q7. A cell gave a resistance of 80Ω when filled with a solution whose specific conductance is $0.020 \Omega^{-1} \text{ cm}^{-1}$. The cell constant is

(a) 0.5 cm^{-1}

(b) 1.6 cm^{-1}

(c) 0.020 cm^{-1}

(d) 1.0 cm^{-1}

$$R = 80 \Omega$$

$$K = 0.020 \Omega^{-1} \text{ cm}^{-1}$$

$$\text{Cell constant} = K \times R$$

$$= 0.020 \times 80 = 1.6 \text{ cm}^{-1}$$

(2)

Q8. The moving-boundary method is principally used to determine

(a) molar conductance

(b) transport numbers

(c) equivalent conductance

(d) limiting ionic mobility

(2)

Q9. If the specific conductance of an electrolytic solution is $0.15 \Omega^{-1} \text{ cm}^{-1}$ and the cell constant is 0.20 cm^{-1} , the conductance (G) of the solution in that cell is:

a) $1.00 \Omega^{-1}$

b) $0.75 \Omega^{-1}$

c) $2.00 \Omega^{-1}$

d) $0.30 \Omega^{-1}$

$$R = G \times \text{cell constant}$$

$$G = \frac{K}{\text{Cell constant}}$$

$$= \frac{0.15}{0.20} = 0.75 \Omega^{-1}$$

(2)

Q10. Which of the following factors *does not* affect conductivity of an electrolyte solution?

- (a) size of ions
- (b) degree of hydration of ions
- (c) total current passed
- (d) viscosity of solvent

(1)

Q11. Which statement is correct regarding equivalent conductance (Λ_{eq}) when the electrolyte is diluted?

- (a) Λ_{eq} increases for strong electrolytes
- (b) Λ_{eq} decreases for weak electrolytes
- (c) Λ_{eq} remains constant for all electrolytes
- (d) Λ_{eq} is independent of dilution

(1)

Q12. The value of the equivalent conductance of KCl is given; that of NaCl is given; using Kohlrausch's law you can calculate that of NaNO_3 . This is an example of:

- (a) transport number law
- (b) Kohlrausch's law of independent migration of ions
- (c) Faraday's law
- (d) Ohm's law

(1)